THE SHORT EXACT SEQUENCE OF SHEAVES OF DIFFERENTIALS ON THE TRIPLE CATEGORY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file cotangent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement includes the displayed definitions of Omega_1, Omega_2 and Omega_3 in a matrix environment; all of it is inside the counted statement span.

Editorial note. The three references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the retained preamble of this chapter (sources/stacks-cotangent/retained-preamble.tex, lines 1-63), containing the Xy-pic setup, the comment environments, the theorem declarations and the macros including \NL.

Allowed standard prerequisites (editorial).

- The split short exact sequence of differentials for a composition of ring maps. Algebra, Lemma algebra-lemma-ses-formally-smooth, used at proof lines 1413-1419. Produces the sequence evaluated at an object U.
- Base change for differentials. Algebra, Lemma algebra-lemma-differentials-base-change, used at proof lines 1430-1432. Identifies the third term of the sequence with the value of g_3^{-1}Omega_3.
- The description of the pullback functors g_i^{-1} by precomposition with u_i. used at proof lines 1413-1414. Makes the comparison with the three displayed sheaves possible.
- The definition of the three sheaves Omega_1, Omega_2, Omega_3 in the statement. used at proof lines 1391-1401. Fix the objects being compared.

Source reference. Original lemma 00RV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1017–1017. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 031K: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3987–3987. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original equation 08QT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11071–11071. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:cotangent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *With notation as in (08QT) set*

$$\Omega_1 = \Omega_{\mathcal{O}/A} \otimes_{\mathcal{O}} \underline{B} \text{ on } \mathcal{C}_{B/A}$$
$$\Omega_2 = \Omega_{\mathcal{O}/A} \otimes_{\mathcal{O}} \underline{C} \text{ on } \mathcal{C}_{C/A}$$
$$\Omega_3 = \Omega_{\mathcal{O}/B} \otimes_{\mathcal{O}} \underline{C} \text{ on } \mathcal{C}_{C/B}$$

Then we have a canonical short exact sequence of sheaves of $\underline{C}$-modules

$$0 \to g_1^{-1}\Omega_1 \otimes_{\underline{B}} \underline{C} \to g_2^{-1}\Omega_2 \to g_3^{-1}\Omega_3 \to 0$$

on $\mathcal{C}_{C/B/A}$.

Proof. Recall that g_i^{-1} is gotten by simply precomposing with u_i. Given an object $U = (P \to B, Q \to C)$ we have a split short exact sequence

$$0 \to \Omega_{P/A} \otimes Q \to \Omega_{Q/A} \to \Omega_{Q/P} \to 0$$

for example by Algebra, Lemma 031K. Tensoring with C over Q we obtain a short exact sequence

$$0 \to \Omega_{P/A} \otimes C \to \Omega_{Q/A} \otimes C \to \Omega_{Q/P} \otimes C \to 0$$

We have $\Omega_{P/A} \otimes C = \Omega_{P/A} \otimes B \otimes C$ whence this is the value of $g_1^{-1}\Omega_1 \otimes_{\underline{B}} \underline{C}$ on U. The module $\Omega_{Q/A} \otimes C$ is the value of $g_2^{-1}\Omega_2$ on U. We have $\Omega_{Q/P} \otimes C = \Omega_{Q \otimes_P B/B} \otimes C$ by Algebra, Lemma 00RV hence this is the value of $g_3^{-1}\Omega_3$ on U. Thus the short exact sequence of the lemma comes from assigning to U the last displayed short exact sequence. $\square$